SOFTWARE USER MANUAL

PrideGTS

Integrated GNSS Time Series Analysis Software

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1 Overview

PrideGTS (Pride GNSS Time Series Analysis Software) is open-source GNSS coordinate time series analysis software developed with MATLAB App Designer. It performs the complete processing from multi-source coordinate file input to parameter estimation (Figure 1). abbreviations are listed in the Appendix.

Before use, you only need to know two points: (1) the modules are linked by Hector-compatible **MOM files**, and the N, E and U components of each station are saved as "station_0/_1/_2.mom", respectively; (2) offset epochs are recorded in the MOM file header as "**# offset MJD**" lines, which are read directly by the subsequent outlier removal and Hector computations. All modules support batch processing.

Table 1 PrideGTS functional modules

Module	Access	Function	Input → Output
Data Formatting	Tools menu	Converts PBO (.pos), NGL (.tenv3) and CMONOC (.txt) coordinate files to MOM files; generates MAT structures	Coordinate files → MOM, MAT
Time Series Visualization and Preprocessing	1st tab of the main interface	Displays series, computes missing-data rates, removes outliers, downloads and corrects environmental loading	MOM → MOM
Known Event Import	Tools menu	Writes known offsets from NGL steps.txt or CMONOC file headers to MOM file headers	MOM + event files → MOM headers
Three-Component Offset Detection	Tools menu	Detects unrecorded offsets jointly in N/E/U with the MGLR test	MOM → MOM headers
Missing Data Reconstruction	Tools menu	Reconstructs missing observations with KKF, PCA Iteration, RegEM or VBPCA	MAT → MOM
Hector Module	Tools menu	Offset detection and correction, noise-model selection, parameter estimation, residual generation and PSD estimation	MOM → MOM,CSV,MAT
Common-Mode Error Extraction	2nd tab of the main interface	Extracts and removes CME with PCA, VBPCA, VBRPCA, FastICA or VBICA	MAT → MOM

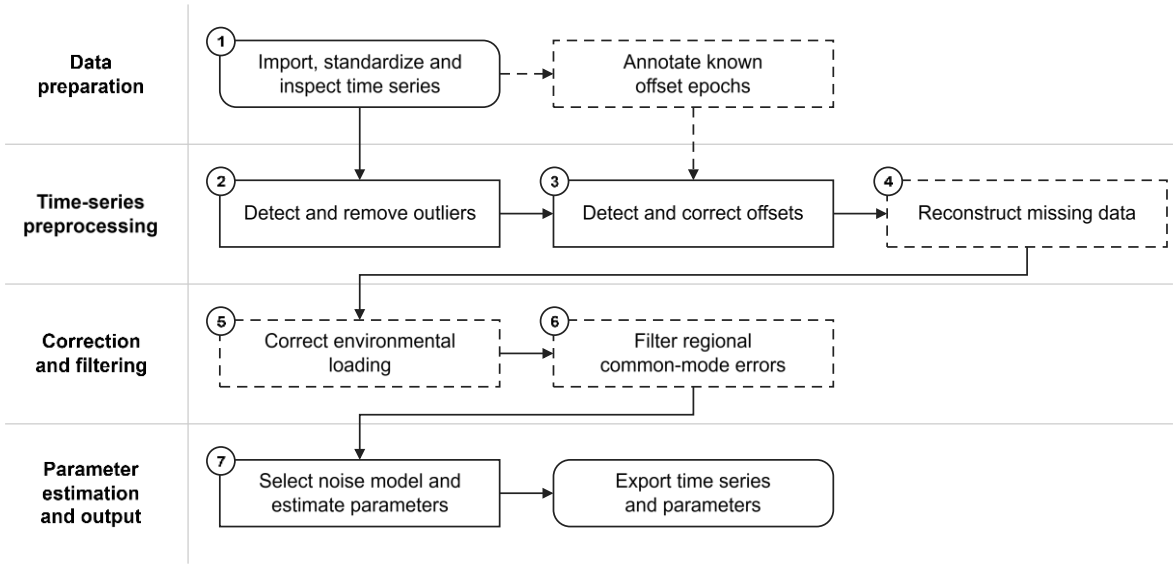


Figure 1 Overview of the workflow of PrideGTS (dashed boxes: optional steps)

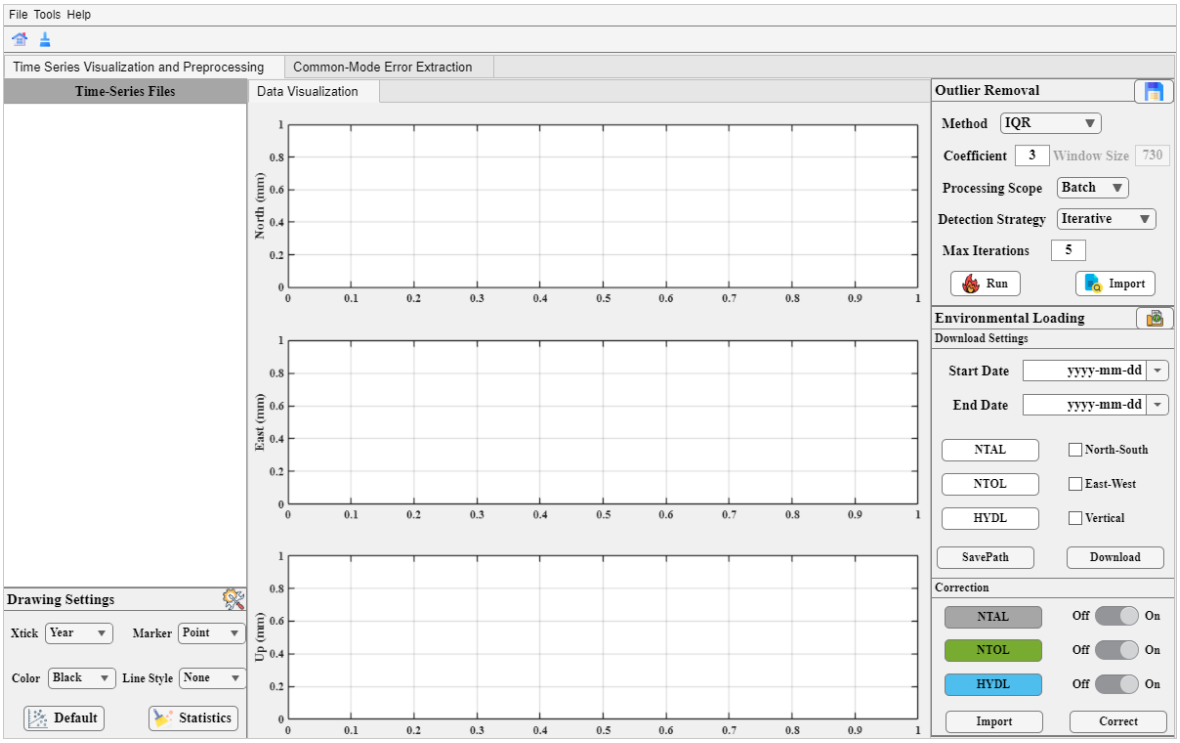


Figure 2 PrideGTS main interface

> **Tip:** For first-time use, please first complete a full run with the sample data following Chapter 3, and then process your own data. Chapter 4 describes the buttons and parameters module by module; if you run into problems, see Chapter 6.

2 Installation

2.1 Install and Launch

1. Download PrideGTS from GitHub (https://github.com/bibinbing/PrideGTS) and extract it to a local directory with write permission;
2. In MATLAB, add the PrideGTS folder and all its subfolders to the path (Home → Set Path → Add with Subfolders);
3. Double-click Install/PrideGTS.mlappinstall and follow the prompts to complete the installation;
4. Click the PrideGTS icon on the APPS tab of MATLAB to launch the software.

To debug the source code: after completing steps 1 and 2, open Install/PrideGTS.mlapp in App Designer and click Run.

2.2 Runtime Environment

Table 2 Software version and runtime environment

Item	Description
Software version	PrideGTS V1.0
MATLAB	R2022b or later (developed under R2022b); only base MATLAB is required, and no additional toolboxes are needed
Operating system	Windows 10/11 (Hector-related functions call Windows executables)
Hector programs	estimatetrend.exe, findoffset.exe and estimatespectrum.exe (Hector 2.1) are shipped with the software in Hector/exe and are called automatically by the software; there is no need to add them to the system PATH

3 Quick Start

This chapter uses the NGL data of 17 Hong Kong stations in sample_data/NGL_HongKong (2014-04-16 to 2024-04-15) to complete one full processing run. Each step gives a “Check” result; if your results agree with it, the step has been performed correctly.

3.1 Preparation

1. Create a working directory Example_HK;
2. Copy the 17 .tenv3 files in sample_data/NGL_HongKong to Example_HK/00_tenv3;
3. Copy sample_data/KnownEvents/steps.txt to Example_HK/00_events.

> **Caution:** Known Event Import, Three-Component Offset Detection and Hector Module modify the MOM file headers directly. Always work on copies; do not process sample_data or original data directly.

Purpose of the other folders in sample_data: PBO_WesternUS (95 PBO .pos files) and CEA_SouthwestChina (51 CMONOC .txt files, 28 of which contain offset records in their headers) can be used to test the other formats; EnvLoading_HongKong provides the NTAL, NTOL and HYDL loading files for the Hong Kong stations.

Table 3 Processing steps of the Hong Kong example

Step	Module/Function	Input	Output
1 Format conversion	Data Formatting:NGL Format	00_tenv3	01_mom
2 Known offset import	Known Event Import	00_events/steps.txt,01_mom	01_mom (headers written)
3 Outlier removal	Outlier Removal	01_mom	02_clean
4 Environmental loading correction	Environmental Loading:Correct	03_envload(mom,NetCDF)	03_envload/corrected
5 Residual generation	Hector:Generate Residual Time Series	03_envload/corrected	corrected/TrendAndSeasonal
6 Missing data reconstruction	Data Formatting:MAT Format;Missing Data Reconstruction:RegEM	TrendAndSeasonal,corrected	04_cme_input,04_res_filled
7 CME filtering	Common-Mode Error Extraction	04_cme_input	05_cme
8 Parameter estimation	Known Event Import;Hector:Parameter Estimation	05_cme	05_cme/estimatetrend

3.2 Step 1: Data Formatting

1. Select **Tools → Data Formatting Module**;
2. Click the Load icon and select 00_tenv3, then click the Save icon and select 01_mom;
3. Keep Coordinate Normalization switched on and click **NGL Format**;

4. In the SDTolerance dialog, keep the default thresholds (N 10 mm, E 10 mm, U 20 mm) and click Confirm;
5. Confirm the save location of ngl.mat; the conversion is complete when "Processing completed" appears.

Check: 51 MOM files (17 stations × 3 components) and ngl.mat are generated in 01_mom. In the main interface, select File → Load Analysis Files (Ctrl+O) to load 01_mom, and click a station to view its N/E/U series; click Statistics to view the missing-data rate of each station (0.41%–9.02%).

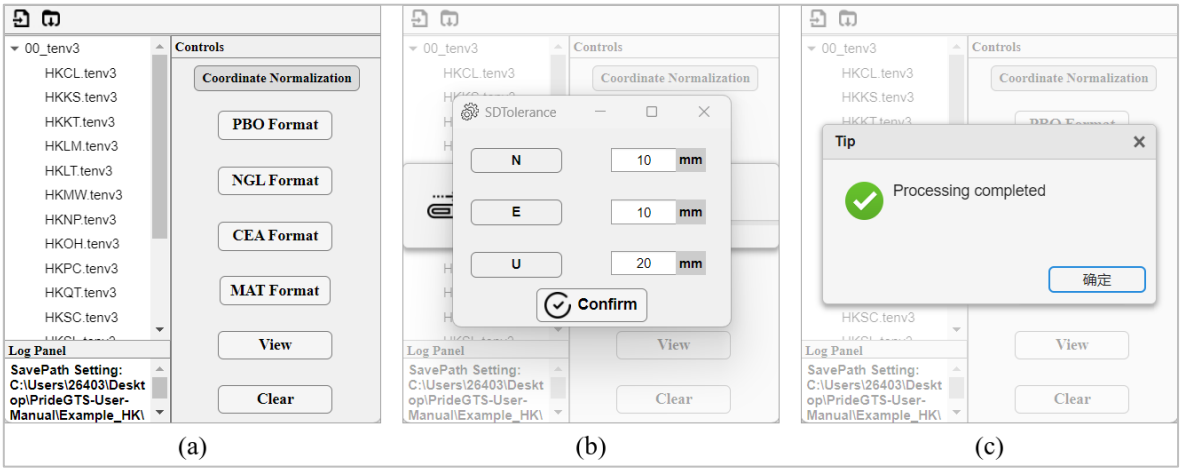


Figure 3 Data Formatting of the Hong Kong data: (a) tenv3 files loaded; (b) SD tolerance setting; (c) conversion completed

3.3 Step 2: Known Event Import

1. Select **Tools → Known Event Import Module**;
2. Set Data Type to NGL Steps File (*.txt), Source Path to 00_events/steps.txt and MOM Data Directory to 01_mom;
3. Click **Analyze Events**;
4. In the Use column, deselect the M7.4 Hualien (Taiwan) earthquake event of 2024-04-02 (MJD 60402) and keep the 13 equipment events;
5. Click **Write Events to MOM Headers**.

Check: 117 events are matched in total, of which 29 lie within the observation period. Reason for deselecting the earthquake event: the epicenter is about 760–810 km from Hong Kong, and most stations have only 5–13 days of observations after the earthquake.

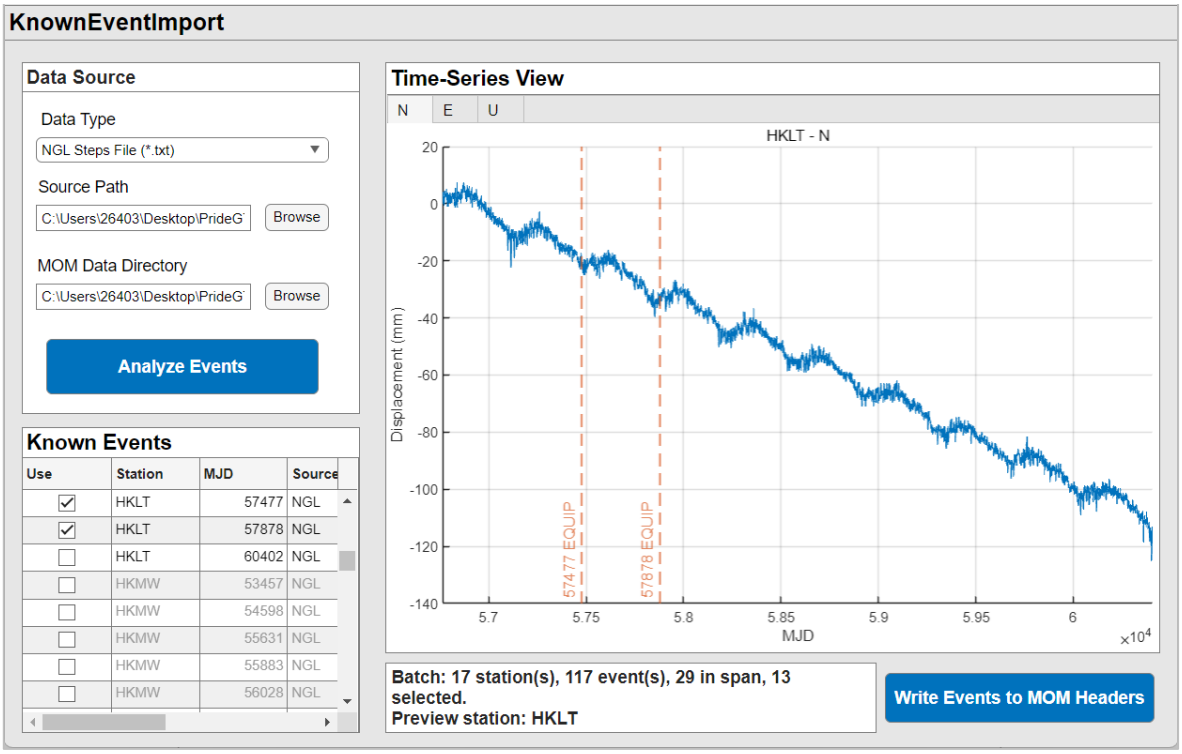


Figure 4 Known Event Import: equipment events of HKLT selected from the NGL steps file

3.4 Step 3: Outlier Removal

1. Return to the main interface, click the save icon in the upper-right corner of the Outlier Removal panel and select 02_clean (it must differ from the input directory);
2. Set Method = IQR, Coefficient = 3, Detection Strategy = Iterative and Max Iterations = 5;
3. Set Processing Scope to Single, select HKLT_0.mom, click **Run**, and check the outliers marked in red;
4. Change Processing Scope to Batch and click **Run** again to process all stations.

Check: 848 observations are removed in total (N 344, E 321, U 183), about 0.47% of the valid observations. The output files keep the offsets in the file headers.

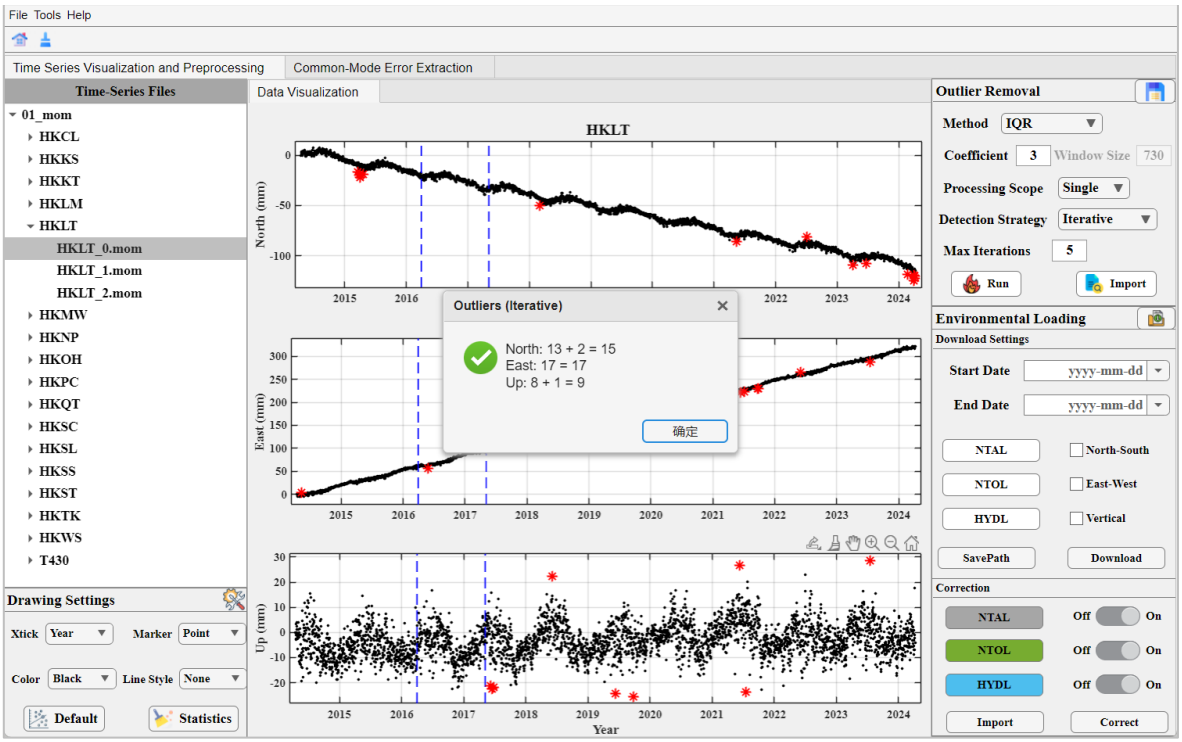


Figure 5 Outlier removal of HKLT (IQR, coefficient 3, iterative)

3.5 Step 4: Environmental Loading Correction

1. Create 03_envload: copy the 51 MOM files in 02_clean to 03_envload/mom, and copy sample_data/EnvLoading_HongKong/NetCDF/merged to 03_envload/NetCDF/merged;
2. In the Correction area, switch on NTAL, NTOL and HYDL;
3. Click Import and select **03_envload** (select the root directory, not mom or NetCDF);
4. Click **Correct**; the results are saved to 03_envload/corrected.

Check: the 17 stations take about 25 s; after removing the linear trend, the vertical standard deviation decreases by about 15% on average. The loading files are already provided with the sample data; when processing your own data, download them first (see 4.3.2).

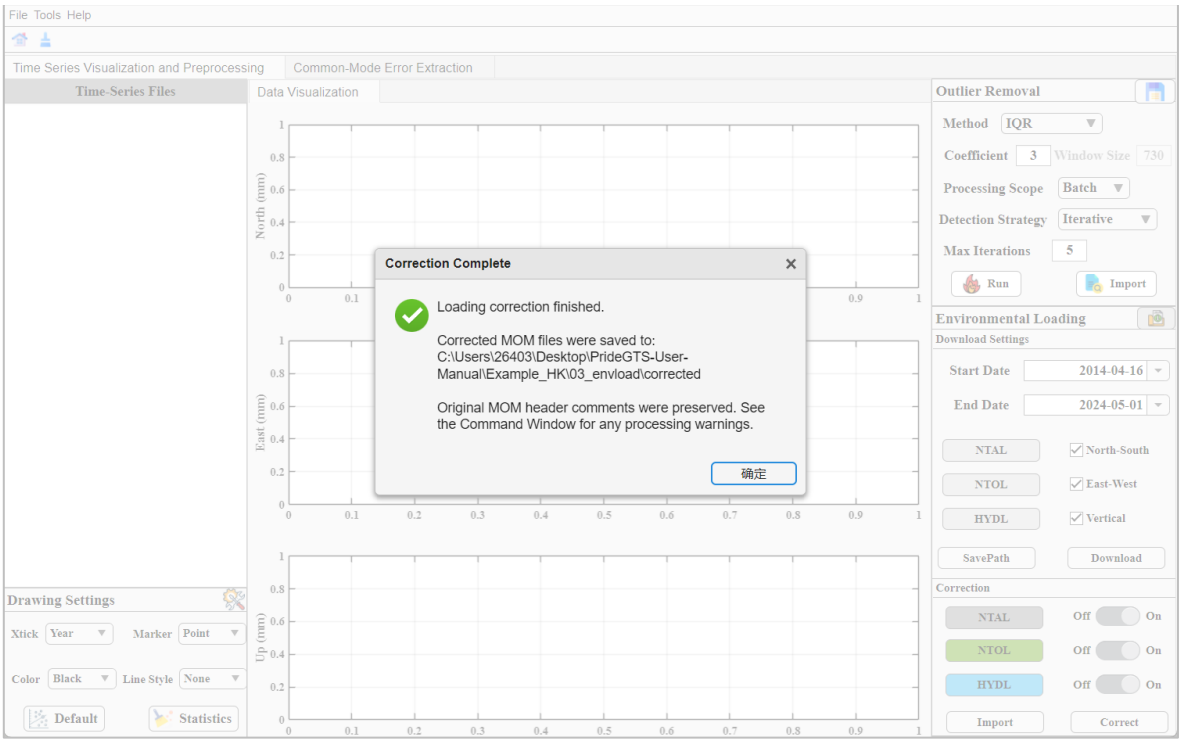


Figure 6 Environmental loading correction with NTAL, NTOL and HYDL

3.6 Step 5: Residual Time Series

1. Select **Tools → Hector Module** and load 03_envload/corrected via File → Load Analysis File;
2. Select **Function → Generate Residual Time Series**;
3. Check Seasonal (annual), Half-seasonal (semi-annual) and Estimate offsets; set Noise mode to Fixed for all files and enter Powerlaw White in Fixed NoiseModels; click OK.

Check: 51 residual MOM files are generated in corrected/TrendAndSeasonal (about 100 s), with the linear trend, annual term, semi-annual term and offsets removed.

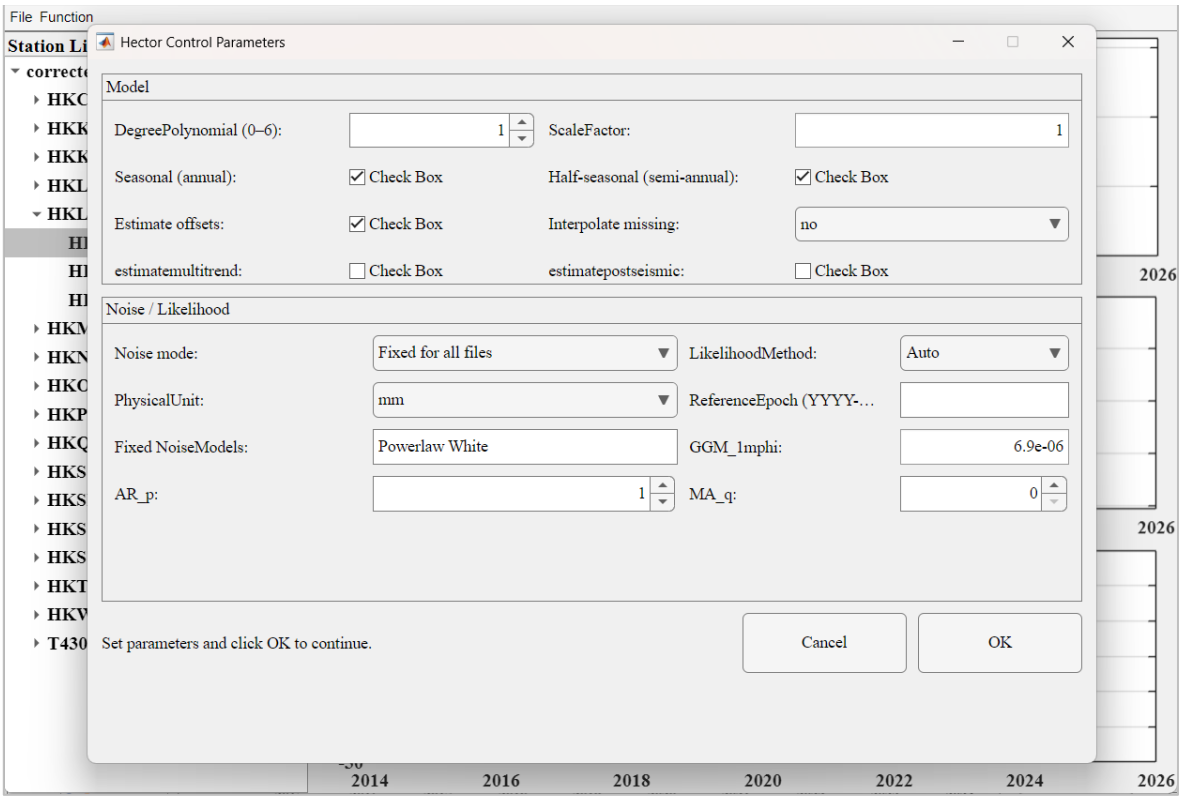


Figure 7 Hector control parameters for generating residual time series

3.7 Step 6: MAT Structures and Missing Data Reconstruction

The CME module requires two MAT files with the same station order and dimensions, and the residuals must not contain missing values (except for VBPCA). Therefore, first generate the MAT files and reconstruct the missing values.

1. In the Data Formatting Module, load corrected/TrendAndSeasonal, click **MAT Format** and save as 04_cme_input/Res_raw.mat;
2. Load corrected, click **MAT Format** and save as 04_cme_input/OriginData.mat;
3. Select **Tools → Missing Data Reconstruction Module**, load Res_raw.mat via File → Load Files, and select **Method → RegEM**;
4. Save the results to 04_res_filled via File → Save Files; then convert them to 04_cme_input/ResData.mat with MAT Format.

Check: both MAT files have 3653 epochs × 17 stations. Click a station to compare the observed and reconstructed values (Figure 8 shows HKMW, the station with the highest missing-data rate). RegEM is used in this example because PCA Iteration produced abnormal reconstructed values of about 70 mm in the long gaps at HKMW.

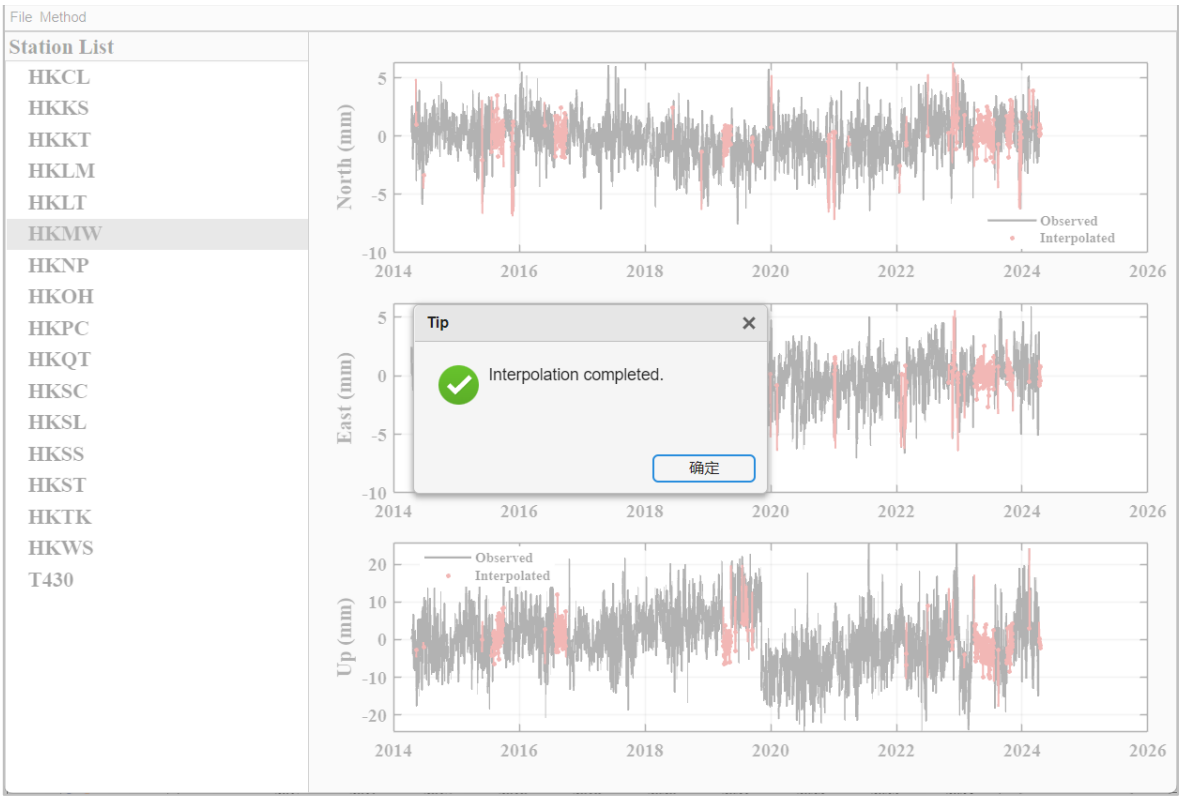


Figure 8 RegEM reconstruction of the residual gaps at HKMW

3.8 Step 7: CME Filtering

1. In the main interface, switch to the **Common-Mode Error Extraction** tab;
2. Click the mat buttons to the right of Original Data and Residual Data to import OriginData.mat and ResData.mat, respectively;
3. Set Select Filter = PCA, PCs = 1 and RefStation = 5 (HKLT);
4. Set Component to North, click **Confirm**, then click **Save** and select 05_cme;
5. Change Component to East and then to Up, repeating step 4 each time.

Check: for the Up component, the first principal component contributes 29.5%, clearly more than the second and third principal components (10.4% and 9.3%), and its spatial responses are uniformly distributed; after removing the CME, the vertical residual RMS decreases by 16.9% on average. 05_cme contains 51 MOM files in total.

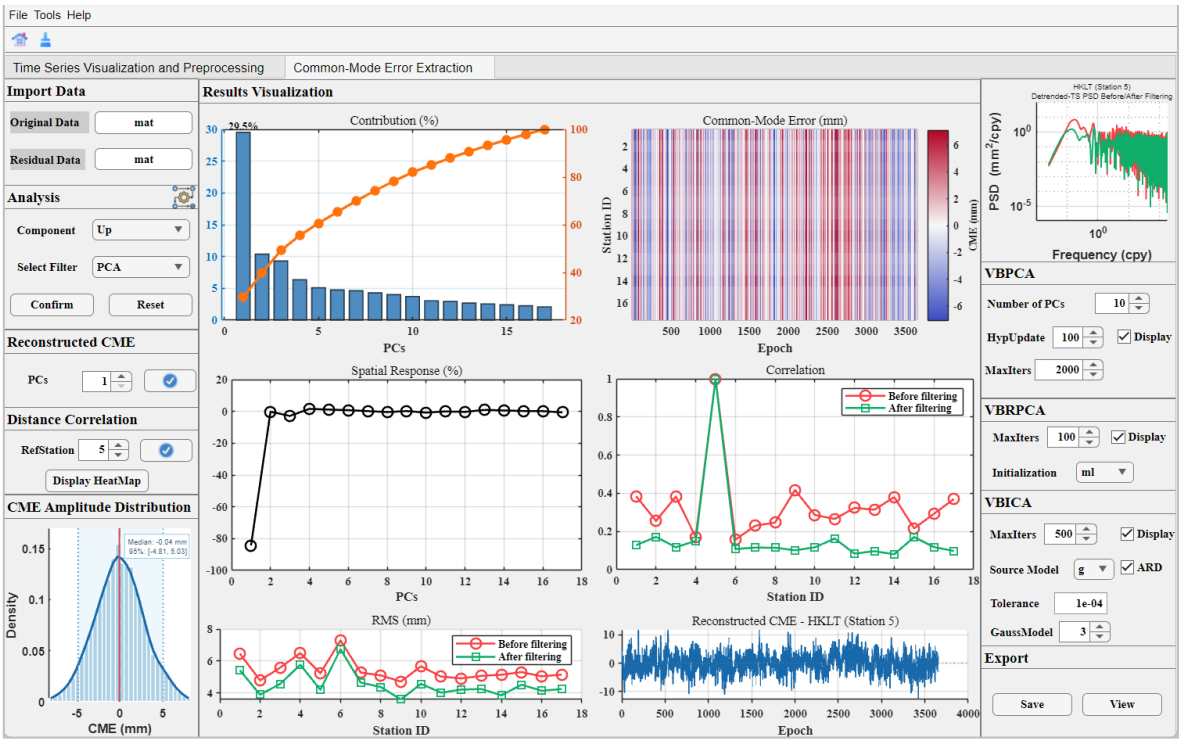


Figure 9 CME extraction of the Up component with PCA (first PC used for reconstruction)

3.9 Step 8: Parameter Estimation

1. The MOM files output by the CME module have no file header: following the steps in Section 3.3, rewrite the 13 equipment events with 05_cme as the MOM Data Directory (repeated writing does not create duplicate offsets);
2. In the Hector Module, load 05_cme via File → Load Analysis File and select **Function → Parameter Estimation**;
3. Set DegreePolynomial = 1 and check Seasonal signal (annual), Half-seasonal (semi-annual) and Estimate offsets; set Noise mode to Fixed for all files and enter FlickerGGM White in Fixed NoiseModels; click OK;
4. In the Parameter Estimation Results window, click **Quick Save CSV**; the results are saved as 05_cme/estimatetrend/station_trends.csv.

Check: the 51 files take about 160 s. The horizontal velocities range from −11.86 to −11.31 mm/yr in the north and from 32.33 to 32.81 mm/yr in the east; the vertical subsidence rates of the reclaimed-land stations HKCL and HKMW are 1.48 mm/yr and 3.41 mm/yr, respectively (Table 4).

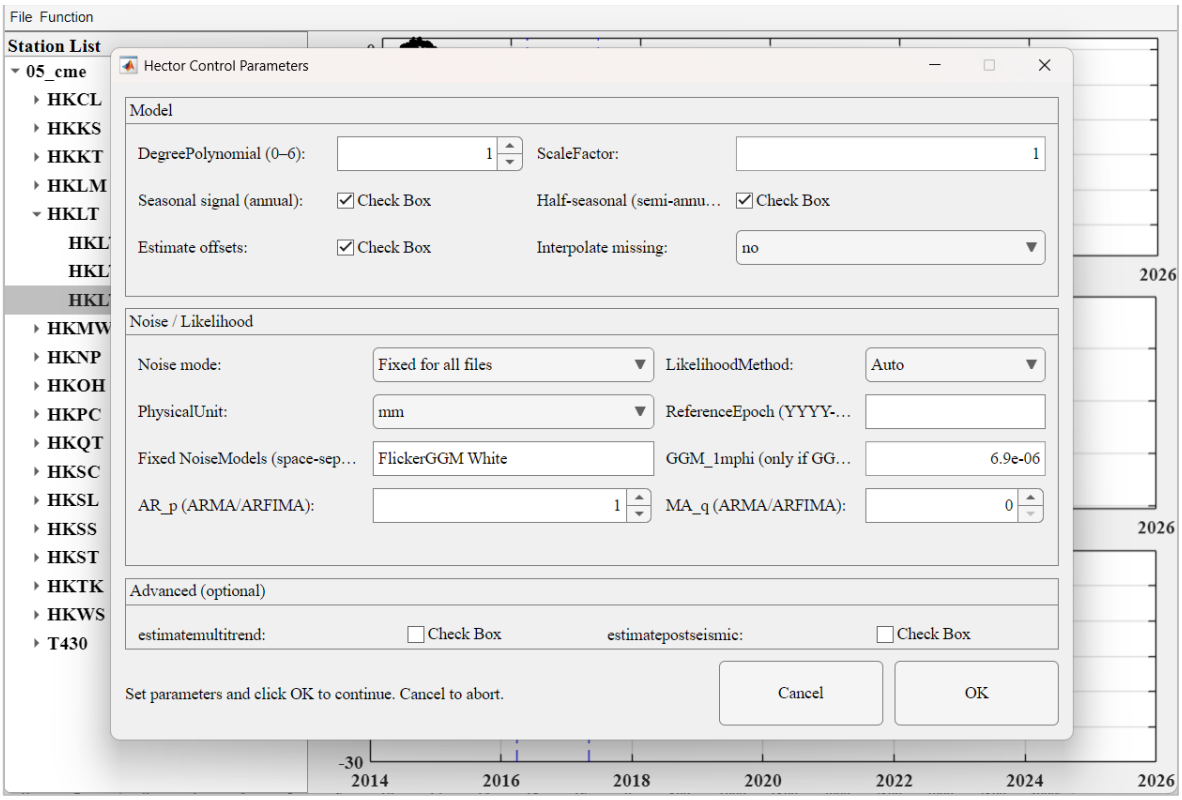


Figure 10 Hector control parameters for parameter estimation

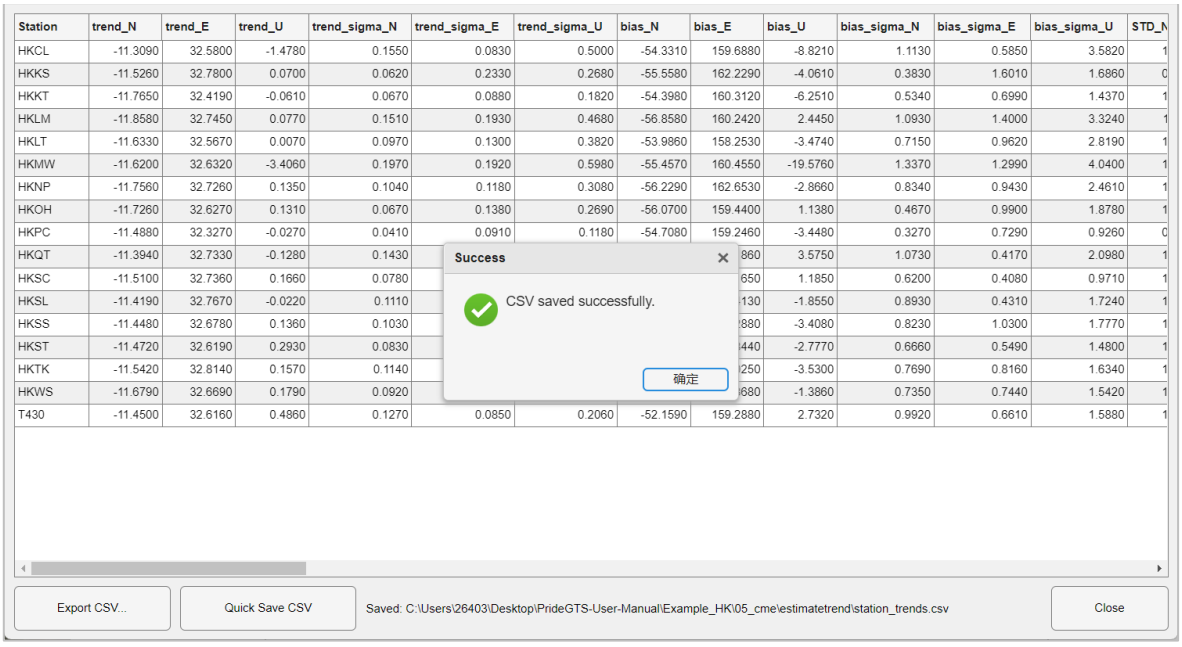


Figure 11 Parameter estimation results of the 17 stations

Table 4 Reference velocity estimates of the Hong Kong stations (FlickerGGM + White noise model)

Station	VN (mm/yr)	VE (mm/yr)	VU (mm/yr)	U annual amplitude (mm)
HKCL	−11.31 ± 0.15	32.58 ± 0.08	−1.48 ± 0.50	2.52
HKKS	−11.53 ± 0.06	32.78 ± 0.23	0.07 ± 0.27	4.66
HKKT	−11.77 ± 0.07	32.42 ± 0.09	−0.06 ± 0.18	3.71
HKLM	−11.86 ± 0.15	32.74 ± 0.19	0.08 ± 0.47	3.09
HKLT	−11.63 ± 0.10	32.57 ± 0.13	0.01 ± 0.38	2.41
HKMW	−11.62 ± 0.20	32.63 ± 0.19	−3.41 ± 0.60	3.12
HKNP	−11.76 ± 0.10	32.73 ± 0.12	0.14 ± 0.31	4.86
HKOH	−11.73 ± 0.07	32.63 ± 0.14	0.13 ± 0.27	3.89
HKPC	−11.49 ± 0.04	32.33 ± 0.09	−0.03 ± 0.12	3.66
HKQT	−11.39 ± 0.14	32.73 ± 0.06	−0.13 ± 0.28	4.82
HKSC	−11.51 ± 0.08	32.74 ± 0.05	0.17 ± 0.12	3.95
HKSL	−11.42 ± 0.11	32.77 ± 0.05	−0.02 ± 0.22	4.96
HKSS	−11.45 ± 0.10	32.68 ± 0.13	0.14 ± 0.22	2.81
HKST	−11.47 ± 0.08	32.62 ± 0.07	0.29 ± 0.19	3.93
HKTK	−11.54 ± 0.11	32.81 ± 0.12	0.16 ± 0.25	4.41
HKWS	−11.68 ± 0.09	32.67 ± 0.09	0.18 ± 0.19	3.82
T430	−11.45 ± 0.13	32.62 ± 0.09	0.49 ± 0.21	5.54

4 Module Reference

This chapter describes the interface and parameters module by module. The numbers in the interface figures correspond one-to-one to the IDs in the tables below.

4.1 Main Interface

The toolbar is located at the top of the main interface (Figure 12), and the menu bar is in the upper-left corner (Figure 13).

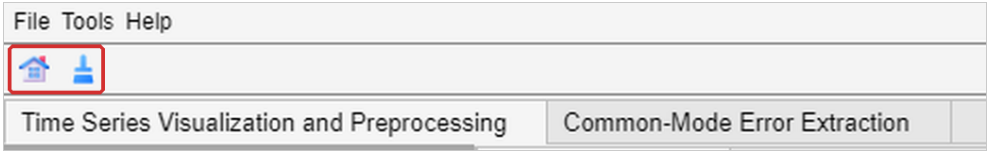


Figure 12 PrideGTS toolbar

Icon	Function
	Restores the default zoom of all plotting windows
	Clears all plots, computation results, loaded files and save paths

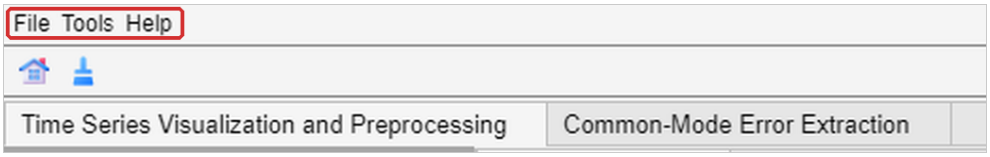


Figure 13 PrideGTS menubar

Fields	Description
File	Load Analysis Files: loads a directory containing MOM files.
Tools	Launches the Data Formatting Module, Missing Data Reconstruction Module, Known Event Import Module, Three-Component Offset Detection and Hector Module.
Help	About: shows developer information; Feedback: opens the PrideGTS GitHub page.

4.2 Data Formatting Module

Input: PBO .pos, NGL .tenv3 (or .tenv3.txt) or CMONOC .txt coordinate files, or a MOM folder; **Output:** three MOM files per station (in mm) and MAT structures.

- PBO format (*.pos): https://www.unavco.org/data/gps-gnss/gps-gnss.html
- NGL format (*.tenv3): https://geodesy.unr.edu/NGLStationPages/GlobalStationList

- CMONOC format (*.txt, published by the First Monitoring and Application Center, China Earthquake Administration):
  https://www.eqdsc.com/eastern/report.shtml?PAGEID=sharedata_station&parid=10&type=sjxl

1. Click Load (ID 1) to select the input folder and click Save (ID 2) to set the output folder (if it is not set, the output is written to the input folder);
2. Click the folder to be converted in the directory tree on the left;
3. Click the format button matching the data (ID 4–6) and set the standard deviation thresholds in the SDTolerance dialog (default N/E/U = 10/10/20 mm; observations exceeding the thresholds are set to NaN);
4. If a MAT structure is needed, load a MOM folder and click MAT Format (ID 7).

A MOM file has two columns (MJD and observation, in mm) and may have a third column of model values; missing observations are NaN. The fields of the MAT structure are day (MJD), site (station name), lat, lon, and dn, de and du (N/E/U series matrices, in mm); the structure is the input to the Missing Data Reconstruction and CME modules.

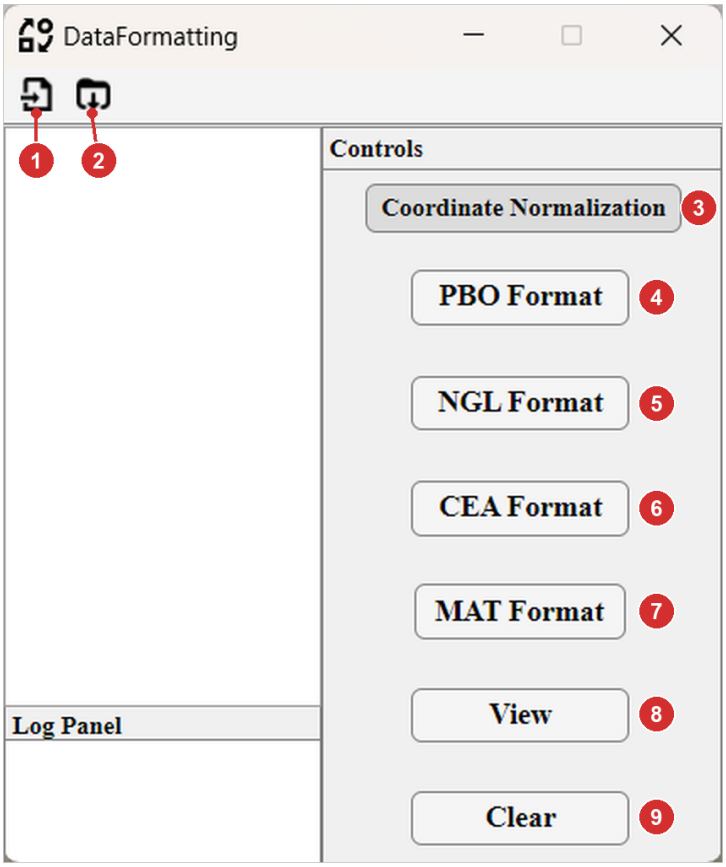


Figure 14 Data Formatting Module

ID	Name	Description
1	**Load Time-Series Files**	Selects the input folder; the folder and its files are shown in the directory tree on the left. If no output folder is set, the input folder also serves as the output folder.
2	**Set Save Folder**	Sets the output folder; no conversion is performed.

ID	Name	Description
3	**Coordinate Normalization**	When on, the value of the first epoch is subtracted from each component; when off, the original coordinates are kept. On by default.
4	**PBO Format**	Converts .pos files: extracts MJD, N/E/U and their standard deviations, converts m to mm, screens the observations by the thresholds, and outputs MOM files and pbo.mat.
5	**NGL Format**	Converts .tenv3 or .tenv3.txt files: extracts MJD, N/E/U, standard deviations and station coordinates, converts m to mm, and outputs MOM files and ngl.mat.
6	**CEA Format**	Converts CEA coordinate time series files (*.txt) and outputs MOM files and cea.mat.
7	**MAT Format**	Merges the three-component MOM files in the folder into a MAT structure (GmisStruct.mat by default) and unifies the time axis of all stations.
8	**View**	Opens the output folder (or the input folder if no output folder is set).
9	**Clear**	Clears the loaded paths and the directory tree; files on disk are not deleted.

> **Note 1:** When Coordinate Normalization is on, the value of the first epoch is subtracted from each component; switch it off if absolute coordinates need to be kept.
>
> **Note 2:** MAT Format generates the structure from MOM files, so lat and lon in the structure are empty. KKF requires station latitudes and longitudes; use pbo.mat, ngl.mat or cea.mat generated by PBO/NGL/CEA Format.
>
> **Note 3:** Multiple folders can be loaded in succession; selecting a directory tree node and pressing Delete only removes it from the interface and does not delete files on disk.

4.3 Time Series Visualization and Preprocessing

Input: a MOM folder (File → Load Analysis Files, Ctrl+O); **Output:** MOM files after outlier removal or loading correction (the original file headers are kept).

Click a station or component in the directory tree to display the N/E/U series of that station in the three plotting areas; Drawing Settings sets the axes and plotting style. Click Statistics to compute the start and end MJD, the number of epochs and the N/E/U missing-data rates of each station.

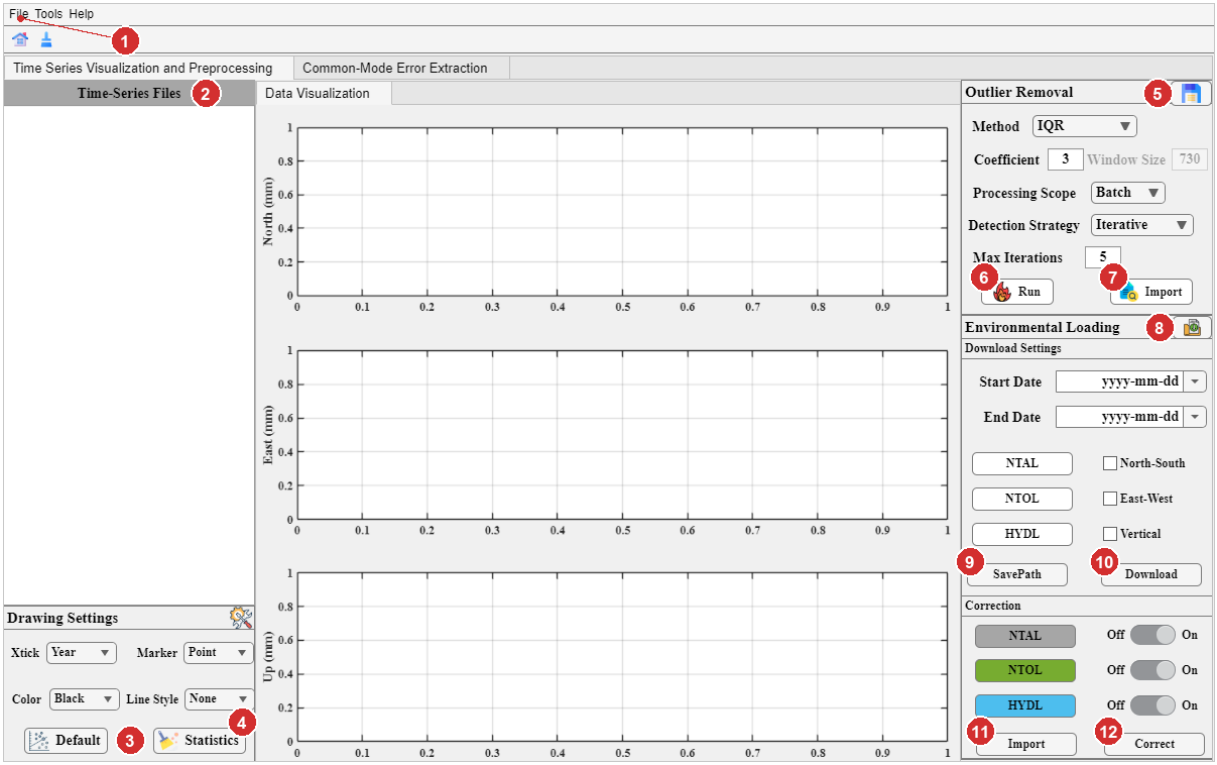


Figure 15 Time Series Visualization and Preprocessing

ID	Name	Description
1	**File**	File → Load Analysis Files (Ctrl+O) loads a directory of MOM files; the files are grouped by station and shown in ID 2.
2	**Time-Series Files**	File tree (directory → station → three-component MOM files). Click a station or component file to display the N/E/U time series of that station.
3	**Default**	Restores the default plotting style (Marker = Point, Color = Black, Line Style = None).
4	**Statistics**	Computes the start and end times, the theoretical number of epochs, the actual number of epochs and the N/E/U missing-data rates of each station.
5	**Outlier Save Path**	Sets the output folder for the outlier removal results; it must be set before Run.
6	**Run**	Runs outlier detection. Single processes the current station and marks the outliers; Batch processes all stations in the directory. Results are saved to the folder set in ID 5.
7	**Import**	Selects another MOM input folder (which must differ from the output folder) and processes it with the current parameters immediately after loading.
8	**Import Station Information**	Imports the station information file (.txt), which contains at least the station name, latitude and longitude, with optional start and end dates; if no dates are given, Start Date and End Date are used.
9	**SavePath**	Sets the root directory for saving the environmental loading data.
10	**Download**	Downloads loading data by station, period, product and direction; select at least one product and one direction.
11	**Import (Correction)**	Selects the working root directory for the correction; this directory must contain a mom subdirectory and a NetCDF (or legacy data) subdirectory.
12	**Correct**	Performs the environmental loading correction; results are saved to the corrected folder.

4.3.1 Outlier Removal

For each component, “bias + linear trend + annual term + semi-annual term + offsets in the file header” is fitted, and epochs whose residuals fall outside “median ± k·IQR” (IQR method) or “median ± k·1.4826·MAD” (MAD method) are identified as outliers and set to NaN. WindowIQR and WindowMAD compute the thresholds only within a window of length Window Size centered on the epoch. In Iterative mode, the model is refitted after each round of removal until no new outliers are found or Max Iterations is reached.

Table 5 Outlier removal parameters

Parameter	Default	Description
Method	IQR	IQR, MAD, WindowIQR or WindowMAD
Coefficient	3	Threshold coefficient k; the smaller k is, the more observations are removed
Window Size	730	Window length (number of samples), used only by the Window methods
Processing Scope	Batch	Single: processes only the current station and marks outliers in the plot; Batch: processes all stations
Detection Strategy	Iterative	Single-pass: detects only one round; Iterative: detects iteratively
Max Iterations	5	Maximum number of iterations in Iterative mode

Tip: The output folder (ID 5) must be set before running. Offsets not written to the file header go into the residuals and cause normal observations near the offsets to be misidentified as outliers, so import the known offsets first (4.4).

4.3.2 Environmental Loading Correction

This function downloads and removes the NTAL, NTOL and HYDL displacement products in the CF frame released by ESMGFZ (Dill and Dobslaw 2013).

Download: click Import Station Information (ID 8) to import the station information file, set Start Date and End Date, select the products and directions, set SavePath (ID 9), and then click Download (ID 10). The station information file is a .txt file without a header line, with one station per line and Tab-separated fields: station name, latitude, longitude, and optional start and end dates, e.g. “BJ01 39.63 115.95 2024-01-01 2024-01-06”.

Correction: select the products in the Correction area, click Import (ID 11) to select the working root directory, and then click Correct (ID 12). The software averages the products with a sampling interval shorter than 24 h (NTAL and NTOL) to daily means, matches them to the MOM epochs by date, converts them to mm and subtracts them from N, E and U; the results are saved to the corrected folder. Epochs without a matching date give NaN.

Directory requirement: The working root directory must contain both the mom and NetCDF subfolders (see the figure below). Downloaded files are saved in NetCDF/parts (segments) and NetCDF/merged (merged by station and product, e.g. BJ01_HYDL.nc).



4.4 Known Event Import Module

Input: NGL steps.txt or a folder of CMONOC coordinate files, and a MOM folder; **Output:** "# offset MJD" records written to the MOM file headers.

- NGL Steps File: code 1 denotes equipment events (EQUIP), and code 2 denotes earthquake events (EQ, which only indicates that the station lies within NGL's magnitude–distance threshold and does not necessarily produce an offset); the events are written to all three N/E/U files. Latest file: https://geodesy.unr.edu/NGLStationPages/steps.txt;
- CEA Time-Series Folder: offset epochs are read from the "n/e/u component" sections of the CMONOC file headers and written only to the corresponding component.

1. Select the event source in Data Type, and select Source Path and MOM Data Directory via Browse;
2. Click Analyze Events to match the events by station name and observation period; events outside the observation period are marked "outside span";
3. Click a table row to view the event position in the plot on the right, and check the required events in the Use column;
4. Click Write Events to MOM Headers. Existing records are kept, and the same epoch is written only once.

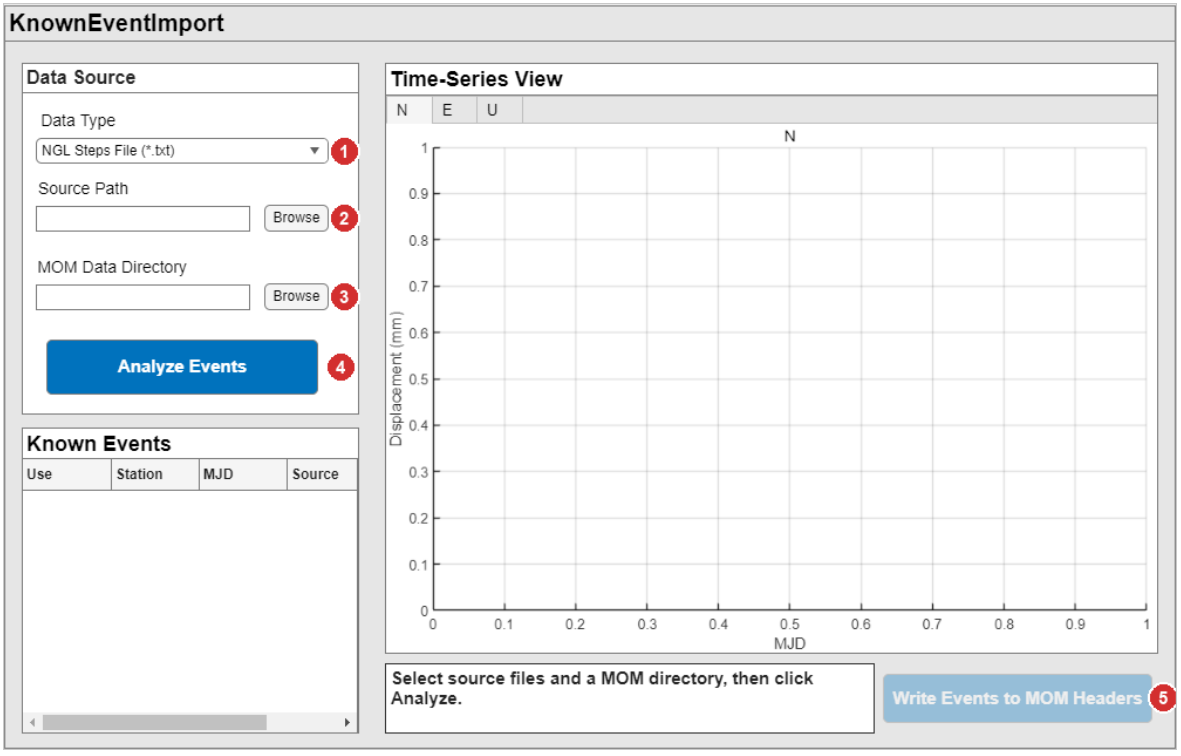


Figure 16 Known Event Import interface

ID	Name	Description
1	**Data Type**	Selects the event source type: NGL Steps File (*.txt) or CEA Time-Series Folder.
2	**Source Path / Browse**	In NGL mode, select one steps.txt file; in CEA mode, select the folder containing the CEA coordinate time series files.
3	**MOM Data Directory / Browse**	Selects the MOM directory to be written. Stations with incomplete three components are skipped; if there is no complete station, the analysis stops with a message.
4	**Analyze Events**	Matches the events by station name and observation period; the results are shown in the Known Events table and the Time-Series View.
5	**Write Events to MOM Headers**	Writes the checked events that lie within the observation period to the file headers as # offset MJD; existing records are kept, and only one record is kept for the same epoch.

4.5 Three-Component Offset Detection Module

Input: a MOM folder (each station must have the three files _0, _1 and _2); **Output:** new offset epochs written to the three MOM file headers.

The multivariate generalized likelihood ratio (MGLR) test (Amiri-Simkooei et al. 2019) is used to detect unrecorded offsets jointly in the N, E and U components. Offsets already in the file header are fitted directly as known terms and are not tested again.

1. Select Tools → Three-Component Offset Detection and load a MOM folder via File → Load Files;
2. Select Analysis → Run Offset Detection to process all stations with complete three components;
3. Click a station in the directory tree and check the offset positions marked by vertical lines.

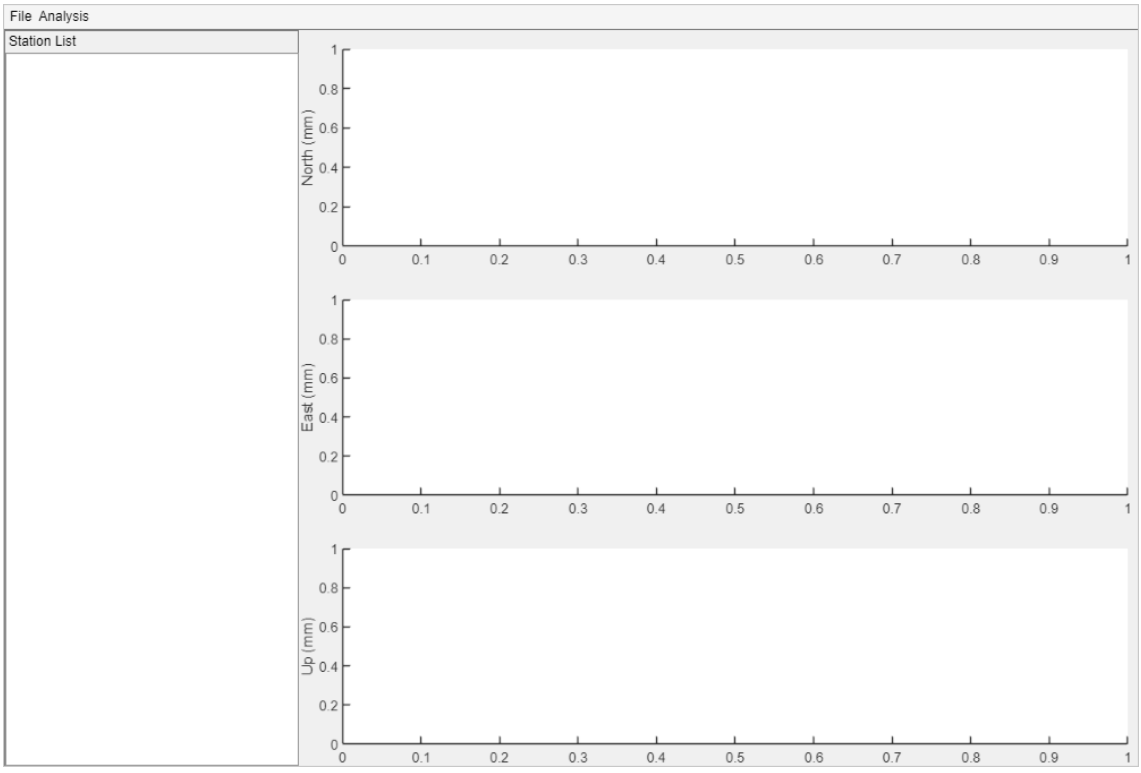


Figure 17 Three-Component Offset Detection (MGLR) interface

> **Note 1:** This module only detects offsets and does not estimate their magnitudes; to estimate and correct the magnitudes, use Apply Offset Corrections in the Hector Module (4.7).
>
> **Note 2:** The test parameters are set in MGLR/code/offset_defaults.m. The defaults are a significance level of 0.001 (station-level Bonferroni correction, Dunn 1961), at least 30 observations on each side of an offset, and white noise + flicker noise as the noise model. α

4.6 Missing Data Reconstruction Module

Input: a MAT structure containing NaN (File → Load Files); **Output:** reconstructed three-component MOM files (File → Save Files, without file headers).

Table 6 Missing data reconstruction methods

Method	Principle	Recommendation
Kriged Kalman Filter	Kriging spatial interpolation + Kalman filtering, adapted from GMIS (Liu et al. 2017)	Requires station latitudes and longitudes; can fill only the missing epochs or output the complete filtered series
PCA Iteration	Short gaps of ≤2 epochs are filled by cubic spline interpolation, and the remaining gaps are reconstructed iteratively with the first three principal components (Liu et al. 2018)	Suitable for networks with clear common variations among stations; beware of long gaps
RegEM	After detrending, missing values are estimated iteratively from the inter-station covariance (Schneider 2001)	Suitable for networks with many stations
VBPCA	Variational Bayesian PCA, which estimates the principal components and the missing values simultaneously (Ilin and Raiko 2010; Li et al. 2020)	Time-consuming when there are many stations and epochs

After selecting a method, click a station name on the left to compare the original observations with the reconstruction results.

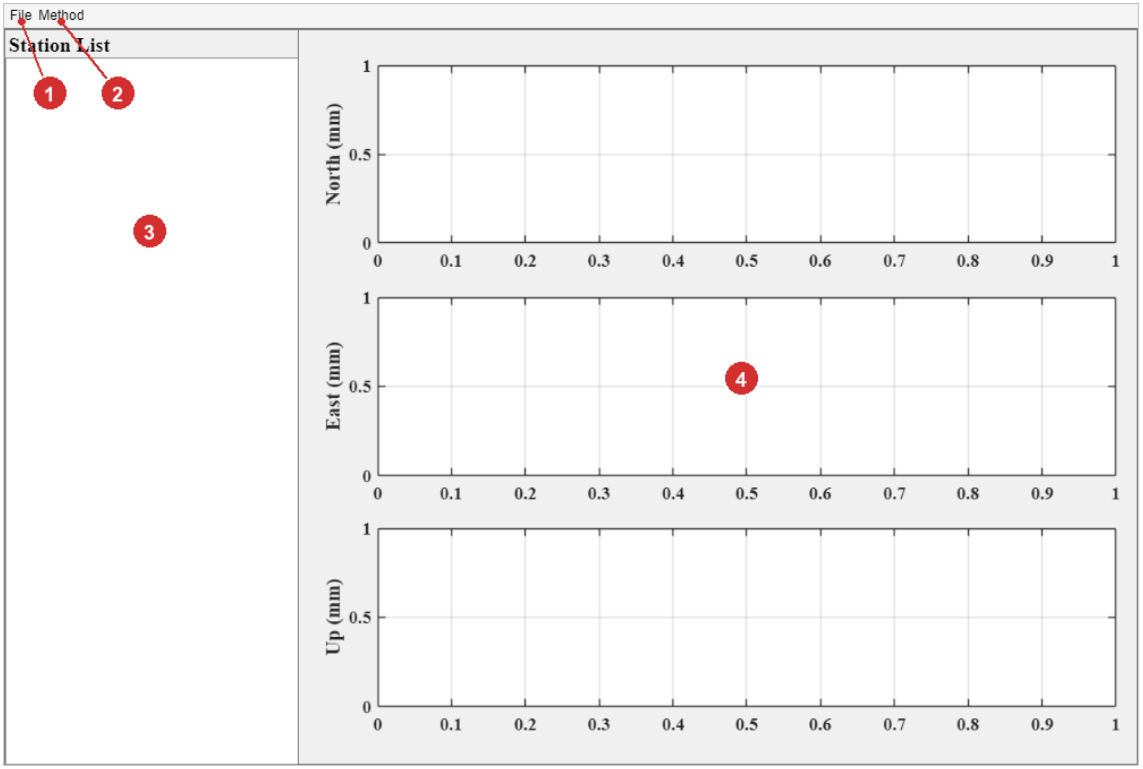


Figure 18 Missing Data Reconstruction interface (Kriged Kalman Filter result)

ID	Name	Description
1	**File**	Load Files imports a MAT file containing a single structure; Save Files exports the results as three-component MOM files for each station.
2	**Method**	Selects the reconstruction method: Kriged Kalman Filter, PCA Iteration, RegEM or VBPCA.
3	**Station List**	Shows all stations; click a station name to switch the current station.
4	**Reconstruction Result View**	Shows the original observations and the reconstruction results of N/E/U for the current station.

> **Caution:** Check the results in long gaps, at the start and end of the series and near offsets carefully; do not mistake smooth results produced by the algorithm for real deformation.

4.7 Hector Module

Input: a MOM folder (File → Load Analysis File); **Output:** see Table 7.

This module calls Hector 2.1 (Bos et al. 2013) in batch mode. After selecting a function in the Function menu, set the parameters in the pop-up Hector Control Parameters window and click OK.

Table 7 Functions of the Hector Module

Function	Purpose	Main outputs
Offset → Find Offset	Detects unrecorded offsets with findoffset; when offsets_3D is on, the three components are detected jointly (an offset is kept only if the total BIC_c decreases)	Written to MOM headers; offset_times.mat
Offset → Apply Offset Corrections	Estimates the magnitude of each offset in the file header with estimatetrend and removes it	corrected folder; OffsetCorrectionLog.csv
Offset → Add Offset Headers	Writes the offsets recorded in offset_times in the MAT file to the MOM file headers	MOM headers
Noise-Model Selection	Fits combinations such as WN, WN+FN, WN+FN+RW, WN+PL and WN+GGM and selects the optimal model by AIC or BIC	AnalysisResults.csv;OptimalNoiseModel.mat
Parameter Estimation	Estimates velocities, annual and semi-annual terms, offsets and noise parameters together with their uncertainties; the noise model can be the optimal model of each file or a uniformly specified model	estimatetrend folder; station_trends.csv
Generate Residual Time Series	Observations minus model values	TrendOnly or TrendAndSeasonal folder
PSD Estimation	Estimates the power spectrum with estimatespectrum, plots the log-log spectrum and overlays the reference slopes of white, flicker and random-walk noise	.out spectrum files; *_psd.png

File → Load Optimal Noise Model (*.mat) reloads an existing OptimalNoiseModel.mat, which provides the per-file models used in Parameter Estimation.

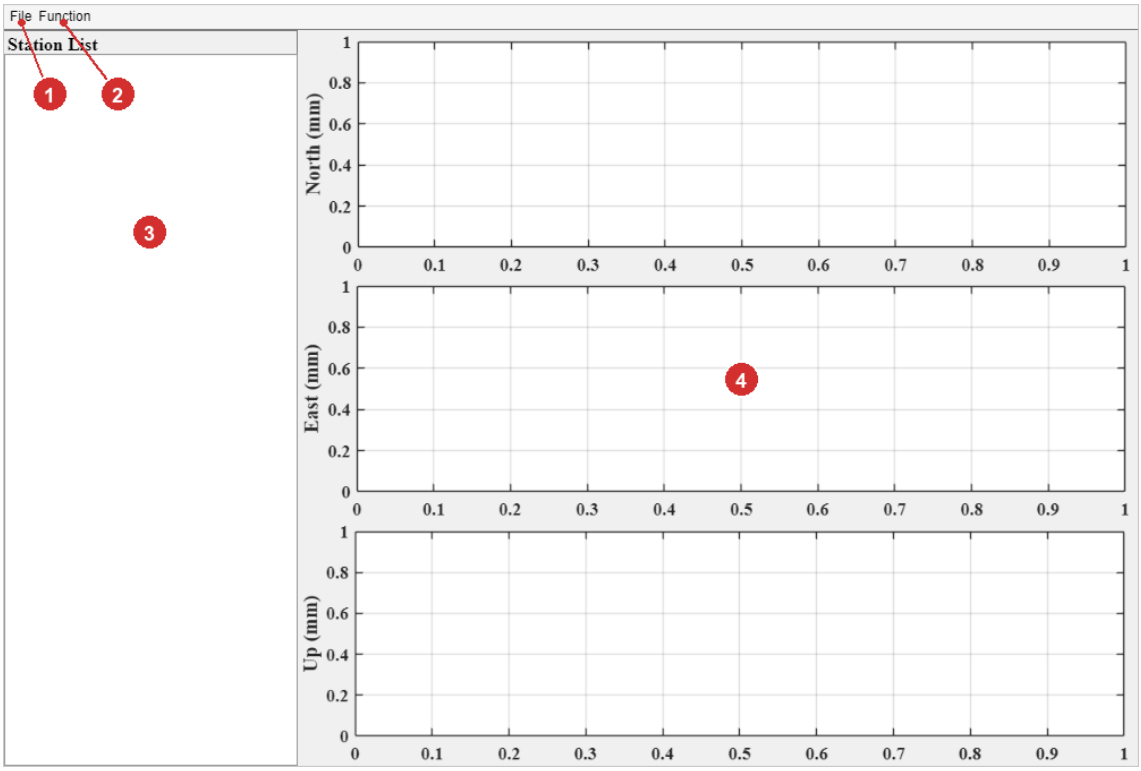


Figure 19 Hector main interface

ID	Name	Description
1	**File**	Loads an analysis directory, or imports an existing optimal noise model.
2	**Function**	Hector computation functions: offsets, noise-model selection, parameter estimation, residual generation and PSD estimation.
3	**Station List**	Shows directories, stations and MOM files; click to switch the current station.
4	**Time-Series Visualization**	Shows the N/E/U time series, the model values and the offset positions in the file header for the current station.

> **Note:** The red curve shows the model values estimated by Hector, and the blue dashed lines show the offset positions in the MOM file header. For the detailed meaning of the Hector parameters, see the Hector 2.1 User Manual.

4.7.1 PSD Estimation

1. Select Function → PSD Estimation;
2. Select one Data File (.mom) or one Data Directory;
3. Modify the parameters in Table 8 as needed and click Run PSD Estimation.

Table 8 PSD Estimation parameters

Parameter	Default	Description
Window Function	Parzen	Window function: Parzen or Hann.
Fraction	0.1	Fraction parameter used by the window function.
Number of Segments	4	Number of segments. Too many segments shorten each segment, and too few reduce the smoothing.
Scale Factor	1	Scale factor of the spectrum.
Output File	estimatespectrum.out	File name of the spectrum data.
Interpolate Missing Data	False	Interpolates missing data. Enable it only when necessary, and check its effect on the high-frequency part.

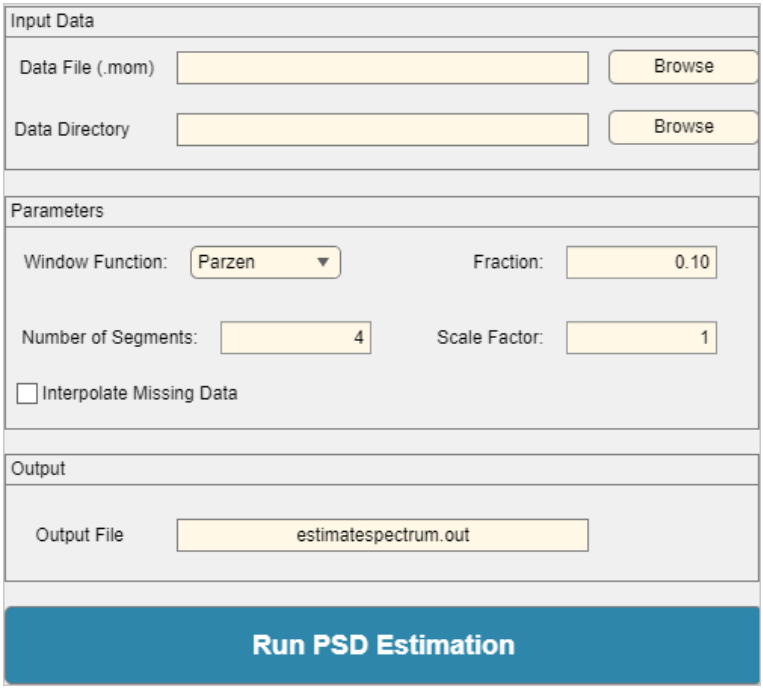


Figure 20 PSD Estimation interface

4.8 Common-Mode Error Extraction Module

Input: Original Data (the MAT file of the coordinate series to be corrected) and Residual Data (the MAT file of the residuals after removing the trend, seasonal terms and offsets); the two must have the same day and site order and the same dimensions; **Output:** MOM files after CME removal, one component per save (without file headers).

Table 9 CME extraction methods

Method	Description	NaN allowed in residuals	Parameter panel (right)
PCA	Principal component analysis (Dong et al. 2006)	No	—
VBPCA	Variational Bayesian principal component analysis (Ilin and Raiko 2010)	Yes	Number of PCs,HypUpdate,MaxIters
VBRPCA	Variational Bayesian robust principal component analysis; determines the rank automatically and separates sparse outliers (Babacan et al. 2012)	No	MaxIters,Initialization
FastICA	Fast independent component analysis (Hyvärinen 1999)	No	—
VBICA	Variational Bayesian independent component analysis (Gualandi et al. 2016)	No	MaxIters,Source Model,Tolerance,ARD,GaussModel

1. Click the mat buttons to the right of Original Data and Residual Data to import the two MAT files;
2. Select Component and Select Filter, and click Confirm;
3. Determine the number of CME components from the contributions, the component time series and the spatial responses, set it in PCs, and click the button on the right to recompute;
4. Click Save to save the current component; the three components must be run and saved separately.

The diagnostic plots include the contributions, the spatial responses, the CME of all stations, the CME and PSD of the reference station (RefStation), the RMS before and after filtering, and the distance correlation between each station and the reference station (distance correlation, ranging from 0 to 1, with the station index on the horizontal axis). If Display HeatMap is pressed before the distance correlation is computed, the inter-station heat map before filtering (lower triangle) and after filtering (upper triangle) is displayed.

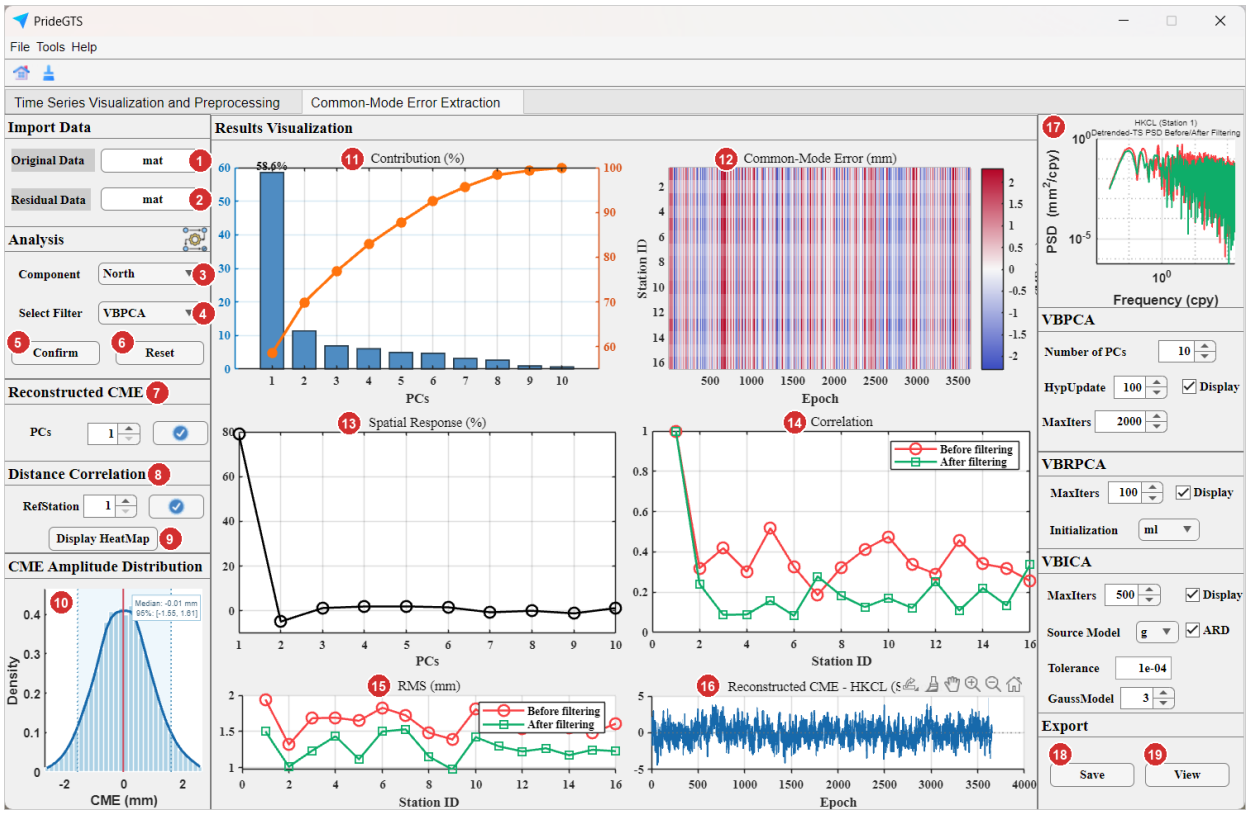


Figure 21 Common-Mode Error Extraction Module interface

ID	Name	Description
1	**Original Data**	Imports the MAT file of the original coordinate time series to be corrected (mm). The day and site order and the matrix dimensions must match those of Residual Data.
2	**Residual Data**	Imports the residual MAT file used for CME extraction (with the trend, seasonal terms and offsets removed).
3	**Component**	Selects the component: North, East or Up.
4	**Select Filter**	Selects the method: PCA, VBPCA, VBRPCA, FastICA or VBICA; the corresponding parameters are set in the panel on the right.
5	**Confirm**	Performs the decomposition and updates all diagnostic plots.
6	**Reset**	Resets Component to North and Select Filter to PCA; data and results are not cleared.
7	**Reconstructed CME / PCs**	Sets the number of components used in the reconstruction; click the button on the right to recompute the CME.
8	**Distance Correlation / RefStation**	Sets the reference station index; click the button on the right to compute the distance correlation between each station and the reference station before and after filtering.
9	**Display HeatMap**	Shows heat maps of the inter-station correlation before and after filtering.

ID	Name	Description
10	**CME Amplitude Distribution**	Probability density distribution of the CME amplitudes, with the median and the 95% interval marked.
11	**Contribution (%)**	Bars show the contribution of each component, and the line shows the cumulative contribution.
12	**Common-Mode Error (mm)**	CME of all stations (epochs on the horizontal axis, station index on the vertical axis, color indicating amplitude).
13	**Spatial Response (%)**	Spatial responses of each component.
14	**Correlation**	Distance correlation between each station and the reference station before and after filtering (station index on the horizontal axis).
15	**RMS (mm)**	RMS of the residuals of each station before and after filtering (relative to zero, not RMSE).
16	**Common-Mode Error (mm)**	CME time series of the reference station.
17	**PSD**	Power spectral density of the residuals of the reference station before and after filtering.
18	**Save**	Saves the current component after CME removal (*_0.mom, *_1.mom or *_2.mom).
19	**View**	Opens the most recent output folder.

Caution: Contributions should be compared only within the same method; the signs of PCA and ICA components are not unique, so do not interpret the signs of the spatial responses on their own. The output files have no # offset header, so the offsets must be re-imported before the files are used in Hector.

5 Input and Output

5.1 Input Formats and Naming Conventions

Table 10 Input formats and naming conventions

Data type	Fields or structure	Naming/directory	Main use
MOM	2–3 columns: MJD, observation, optional model value; the header may contain # offset.	station_0/1/2.mom for N/E/U	Common format between modules
MAT	day, site, lat, lon, dn, de, du; in mm (lat and lon are empty when generated by MAT Format)	—	Missing data reconstruction, CME extraction
Station information TXT	Station name, latitude, longitude; optional start and end dates	—	Environmental loading download
Known offset events	NGL steps.txt or CEA coordinate time series files	—	Written to MOM headers
Environmental loading	NetCDF files of NTAL, NTOL and HYDL	Working root directory containing mom and NetCDF (or data); results written to corrected	Environmental loading correction

5.2 Main Outputs

Table 11 Major outputs

Module	Main outputs	Notes
Data Formatting	MOM files; pbo.mat, ngl.mat or cea.mat; MAT Format outputs GmisStruct.mat	Check the units, station names and N/E/U assignment
Visualization and Preprocessing	MOM files after outlier removal or environmental loading correction	The input and output directories of Import must differ
Known Event Import / Three-Component Offset Detection	Offset epochs written to the MOM headers	Modifies the original files directly; back them up first
Missing Data Reconstruction	Reconstructed MOM files	Re-check the results in long gaps, at the series ends and near offsets
Hector: Find Offset	offset_times.mat; temporary directory findoffset_temp	Requires findoffset.exe
Hector: Apply Offset Corrections	corrected directory and OffsetCorrectionLog.csv	Compare the series before and after the correction
Hector: Parameter Estimation	estimatetrend directory and station_trends.csv	Results depend on the noise model and offset settings
Hector: Noise-Model Selection	OptimalNoiseCombination directory, AnalysisResults.csv, OptimalNoiseModel.mat	Record the criterion used (AIC or BIC)
Hector: Generate Residual Time Series	TrendOnly or TrendAndSeasonal directory	The directory name indicates the deterministic terms that were removed

Module	Main outputs	Notes
Hector: PSD Estimation	estimatespectrum.ctl, .out files and *_psd.png	Requires estimatespectrum.exe
CME Extraction	*_0.mom, *_1.mom or *_2.mom of the current component	Only one component is saved at a time; the output has no file header

6 FAQ

Problem	Solution
Hector-related functions fail to run?	Check whether estimatetrend.exe, findoffset.exe and estimatespectrum.exe exist in Hector/exe, whether they are blocked by Windows or antivirus software, and whether the working directory is writable.
Import in outlier removal reports an error?	The input folder selected in Import must differ from the output folder (Outlier Save Path).
Environmental loading correction cannot find the data, or all results are NaN?	Import should select the working root directory that contains both mom and NetCDF; check whether the period of the loading files covers the MOM epochs and whether the corresponding products are selected.
KKF reports missing latitudes and longitudes?	The structure generated by MAT Format contains no latitudes or longitudes. Use pbo.mat, ngl.mat or cea.mat generated by PBO/NGL/CEA Format instead.
The CME module reports an error or gives abnormal results?	Original Data and Residual Data must have the same station order and dimensions; it is recommended to generate both from the same set of MOM files with MAT Format. Except for VBPCA, the residuals must not contain NaN, so perform missing data reconstruction first.
Offsets are not estimated in parameter estimation?	The MOM files output by the CME module and Missing Data Reconstruction have no file headers; rewrite the known offsets with Known Event Import (the results of Hector Find Offset can be written back with Add Offset Headers).
How to recover after a processing error?	Known Event Import, Three-Component Offset Detection and Find Offset modify the MOM files directly; back up the MOM folder before processing and restore it from the backup if an error occurs.

7 Contact

PrideGTS is currently in its first open-source release. If you encounter problems, please send an e-mail including the data format, processing steps and error messages.

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Appendix Abbreviations

The abbreviations used in this manual are listed below (in alphabetical order).

Abbreviation	Full name
AIC	Akaike Information Criterion
ARD	Automatic Relevance Determination
BIC	Bayesian Information Criterion
BIC_c	Corrected Bayesian Information Criterion
CEA	First Monitoring and Application Center, China Earthquake Administration
CF	Center of Figure
CME	Common-Mode Error
CMONOC	Crustal Movement Observation Network of China
ESMGFZ	Earth System Modelling Group at GFZ
FastICA	Fast Independent Component Analysis
FN	Flicker Noise
GGM	Generalized Gauss–Markov (noise)
GNSS	Global Navigation Satellite System
HYDL	Hydrological Loading
ICA	Independent Component Analysis
IQR	Interquartile Range
KKF	Kriged Kalman Filter
MAD	Median Absolute Deviation
MAT	MATLAB Data File (*.mat)
MGLR	Multivariate Generalized Likelihood Ratio
MJD	Modified Julian Date
MOM	Time Series File Format of Hector (*.mom)
N/E/U	North/East/Up
NaN	Not a Number
NGL	Nevada Geodetic Laboratory
NTAL	Non-Tidal Atmospheric Loading
NTOL	Non-Tidal Oceanic Loading
PBO	Plate Boundary Observatory
PCA	Principal Component Analysis
PL	Power-Law Noise
PSD	Power Spectral Density

Abbreviation	Full name
RegEM	Regularized Expectation Maximization
RMS	Root Mean Square
RMSE	Root Mean Square Error
RW	Random-Walk Noise
SD	Standard Deviation
VBICA	Variational Bayesian Independent Component Analysis
VBPCA	Variational Bayesian Principal Component Analysis
VBRPCA	Variational Bayesian Robust Principal Component Analysis
WN	White Noise